

Sybil Attacks on Inclusion Lists

FOCIL, AUCIL and Beyond

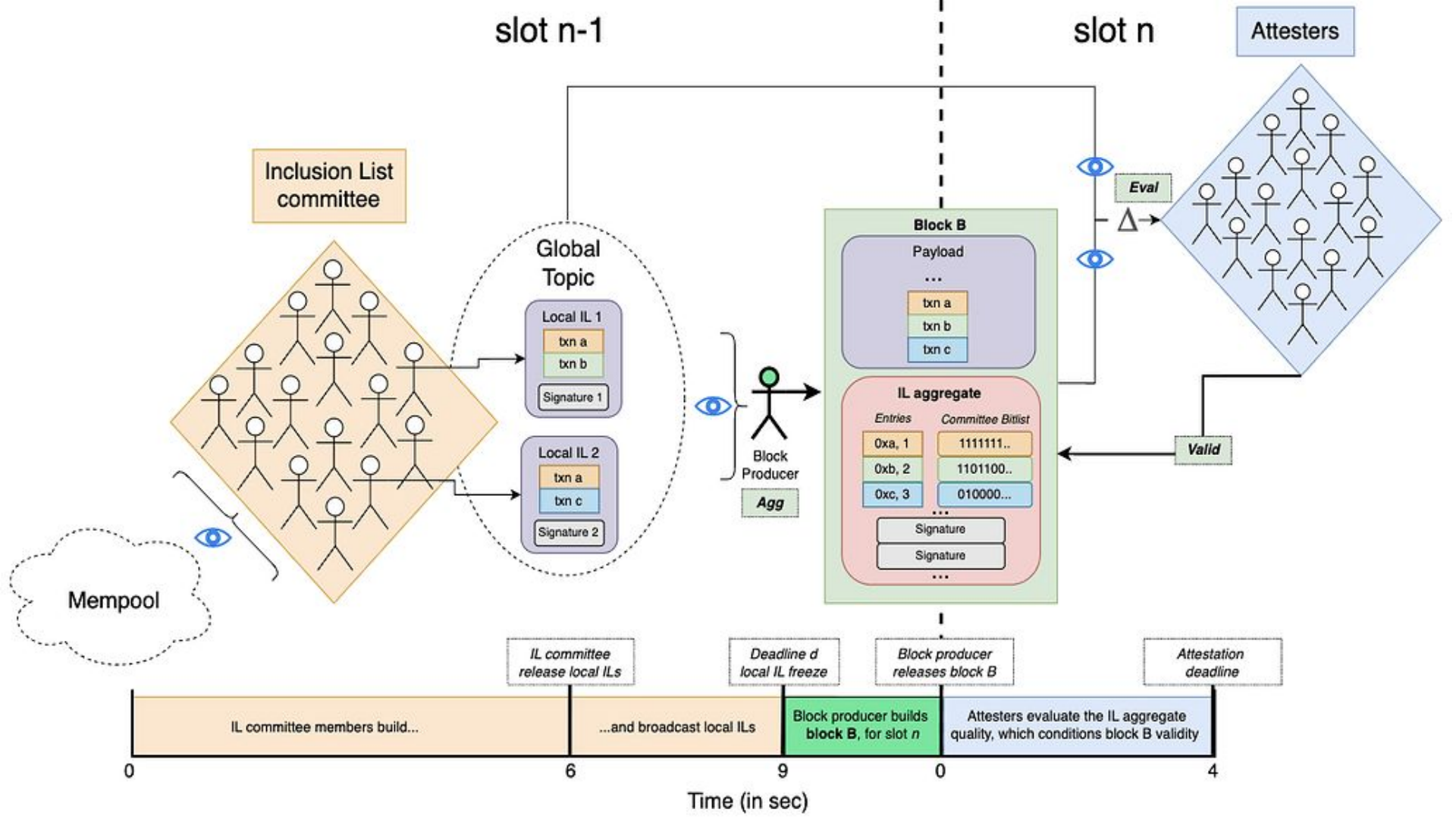
Abhimanyu Nag

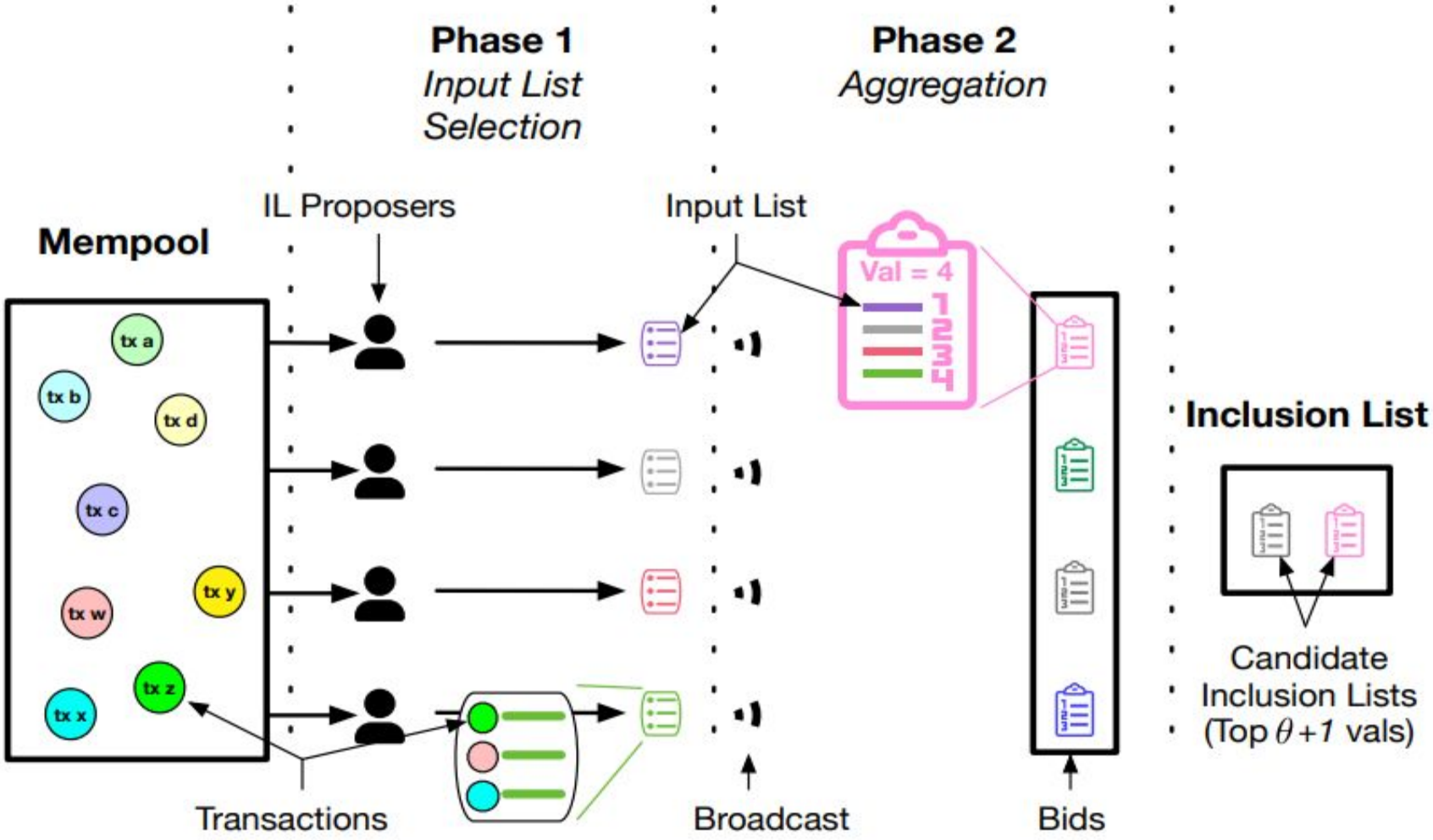
MITACS Fellow, University of Alberta


INCLUSION LIST MECHANISMS
are streaming into you

WHAT ARE THEY?

- Method of txn inclusion by proposers under PBS. Censorship Resistance galore
- Fork-Choice Enforced Inclusion Lists (FOCIL)
  - Consensus level, no pricing
- Auction Based Inclusion Lists (AUCIL)
  - Economic level, price formation


slot n-1
slot n
Attesters
Inclusion List committee
Global Topic
Local IL 1
txn a
txn b
Signature 1
Local IL 2
txn a
txn c
Signature 2
Mempool
Block Producer
Agg
Block B
Payload
...
txn a
txn b
txn c
IL aggregate
Entries
Committee Bitlist
0xa, 1
1111111..
0xb, 2
1101100..
0xc, 3
010000...
...
Signature
Signature
...
Eval
Δ
Valid
IL committee release local ILs
Deadline d local IL freeze
Block producer releases block B
Attestation deadline
IL committee members build...
...and broadcast local ILs
Block producer builds block B, for slot n
Attesters evaluate the IL aggregate quality, which conditions block B validity
0
6
9
0
4
Time (in sec)


Phase 1
Input List
Selection
Phase 2
Aggregation
IL Proposers
Input List
Mempool
tx a
tx b
tx d
tx c
tx y
tx w
tx z
tx x
Val = 4
1
2
3
4
Inclusion List
Candidate
Inclusion Lists
(Top θ+1 vals)
Transactions
Broadcast
Bids

FOCIL IS SYBIL PROOF BUT NOT BRIBERY PROOF

Definition 19. *Let Fee_t be the total fee per unit of size that a transaction t gives to the includers and the block producer when it is included in both an inclusion list and a block. The structure of this fee depends on the specifics of the bid b_t and the fee mechanism under examination.*

- *If B_k meets protocol's predetermined criteria for inclusion lists:*

$$
\begin{aligned}
u_{BP}(b_t, H, B_k, \alpha) = & \sum_{t \in B_k \cap \neg(F_{BP} \cup F_{BP}^{Init})} [(p_t^{BP}(H, B_k, \alpha) \\
& - \mu_{BP}^{Cost}) \cdot s_t] - \sum_{t \in B_k \cap M_0} [\mu_{BP}^{Bribe}(s_t, b_t, p_t, \vec{\alpha}_t, H, M_0)] \\
& - \sum_{t \in B_k \cap (F_{BP}^{Init} \cup F_{BP})} [(q_t(H, B_k) + \sum_{i \in \{1,\dots,m\}} p_t^{CM}(H, B_k, \alpha, i))] \cdot s_t \\
& - \sum_{t \in F_{BP}^{Init}} [\gamma \cdot Fee_t \cdot s_t]
\end{aligned}
$$

- *Otherwise: The same with the difference that the first and the third term are equal to 0.*

Source: Stouka, A. P., Ma, J., & Thiery, T. (2025). Multiple proposer transaction fee mechanism design: Robust incentives against censorship and bribery. arXiv preprint arXiv:2505.13751.

AUCILIS NOT SYBIL PROOF BUT IT IS BOUNDEDLY BRIBERY PROOF

Lemma 12. *Given all IL proposers follow strategy Ψ as described in Algorithm 1 in absence of adversary, given $\hat{n}_t$ is the number of input lists that contain the target transaction, and η is the total number of input lists available. If an adversary wants to censor the target transaction (with 100% probability) by bribing during the aggregation phase, then the total cost incurred by the adversary is at least $(n-\theta)\cdot u_{agg}/2$.*

Source: Wadhwa, S., Ma, J., Thiery, T., Monnot, B., Zanolini, L., Zhang, F., & Nayak, K. (2025). AUCIL: An Inclusion List Design for Rational Parties. Cryptology ePrint Archive.

SYBIL ATTACKS ON AUCIL



Source: Sybil Attacks on AUCIL by Abhimanyu Nag.
https://functor.network/user/3197/entry/1845

Definition 1 *AUCIL will be considered sybil proof if, for every number $q \geq 1$ of controlled proposers and every coordinated strategy of those identities,*

$$U_A^{(q)} \leq U_A^{(1)}$$

Example:

Consider three proposers and one transaction per list. All lists are available.

- Transactions X and Y pay 10 and 6, shared equally among contributors.
- Initial allocation: P1 → X, P2 → X, P3 → Y. Rewards: 5, 5, 6.
- Alone, P2 would not switch to Y then its reward falls from 5 to 3.
- An attacker controlling P1 and P2 switches P2 to Y and total reward rises from 10 to 13.

Lemma 1 (Phase I Sybil attack) *Suppose h independent proposers and q identities controlled by A are assigned transaction t, which pays utility u_t. The total utility of A from t is*

$$V_q(t) = \frac{qu_t}{1 + \gamma(h + q - 1)}$$

Moreover,

$$V_q(t) - V_1(t) = \frac{u_t(q-1)\big(1 + \gamma(h-1)\big)}{\big(1 + \gamma(h+q-1)\big)(1+\gamma h)} \geq 0,$$

with strict inequality except when $h = 0$ and $\gamma = 1$. Hence AUCIL's fee sharing rule is not sybil proof.
If t is assigned to n_t identities, of which s_t are already controlled by the censor, then a guaranteed Phase I attack costs at most

$$C_{\mathrm{I}}^{\mathrm{Syb}}(t) \leq (n_t - s_t)u_t,$$

since paying each remaining identity u_t dominates its maximum possible reward from t.

The Effect on Bribery Bounds

Lemma 2 (Phase II Sybil attack) *Suppose the censor controls s of the n selected aggregators. Under AUCIL's bribery model, its additional external bribery cost for certain censorship is*

$$C_{\mathrm{II}}^{\mathrm{Syb}}(s) = [n - \theta - s]_+ \frac{u_{\mathrm{agg}}}{2}$$

Therefore,

$$C_{\mathrm{II}}^{0} - C_{\mathrm{II}}^{\mathrm{Syb}}(s) = \min\{s, n - \theta\} \frac{u_{\mathrm{agg}}}{2},$$

where

$$C_{\mathrm{II}}^{0} = (n - \theta) \frac{u_{\mathrm{agg}}}{2}$$

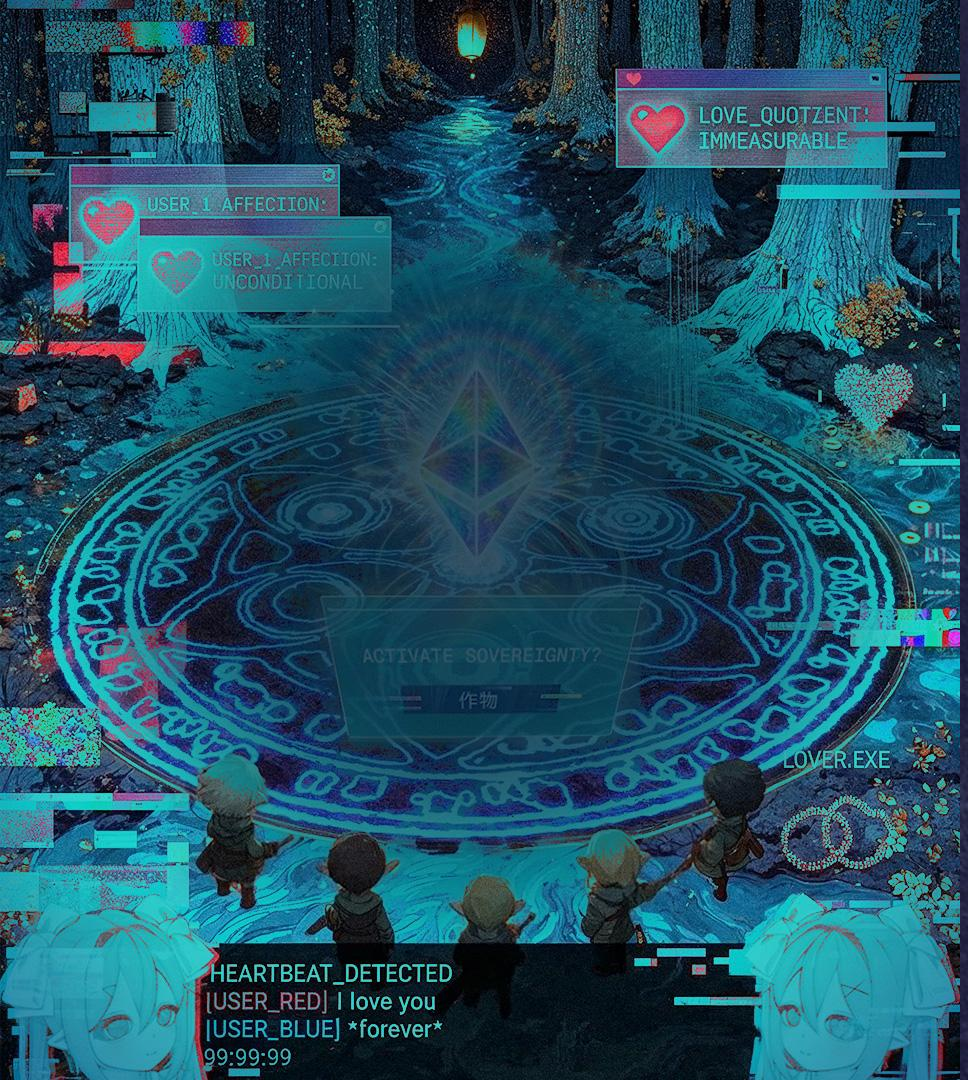
is AUCIL's original aggregation bound.

ABUNDANCE_REGARDLESS_OF_MEANS

- Can we use stake based partitions for fees and improve upon the correlated equilibrium greedy algorithm to optimize for equity?

- Can we improve the scoring auction design in phase II to deal with the Sybil proofness case as well? How about if instead of the VRF sample, we can design a stake weighted lottery?


Thank you!
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@AbhiMan160102



- FOCILIS SYBIL PR

AUCIL



Section 1

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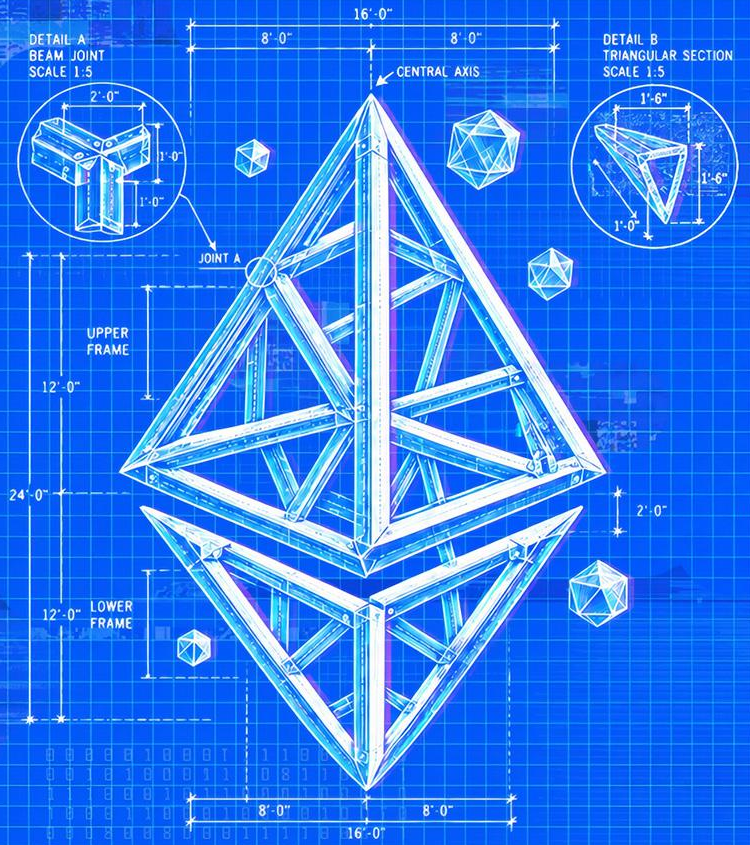
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“Number rules the universe.”
— Pythagoras



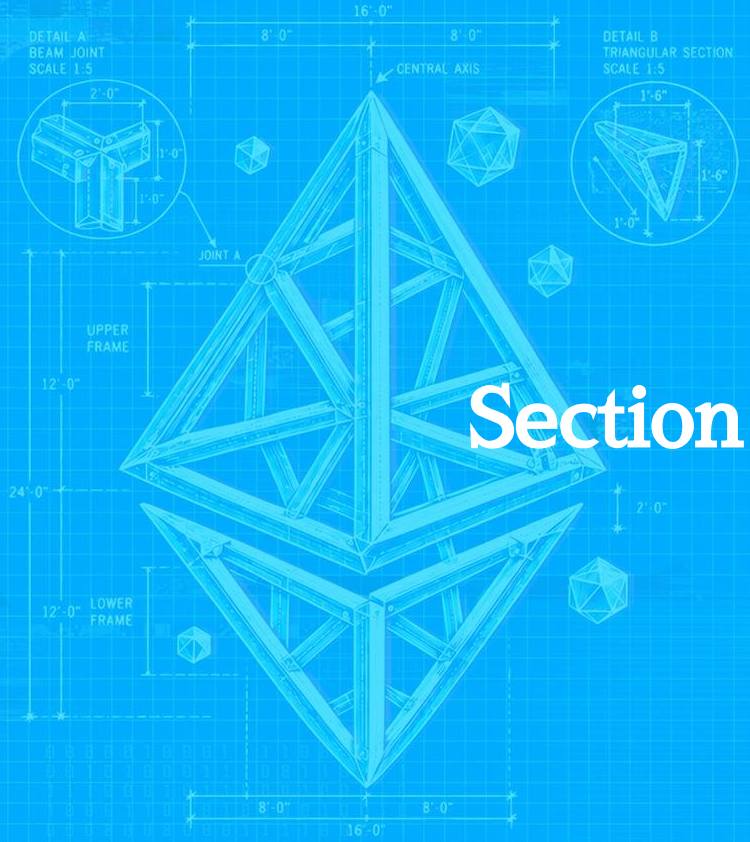
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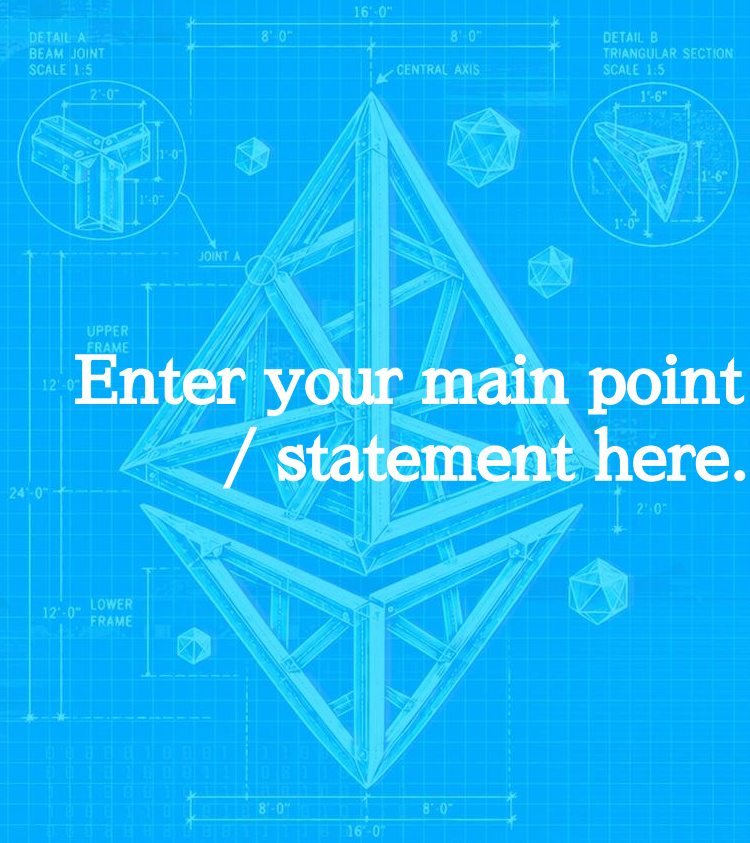
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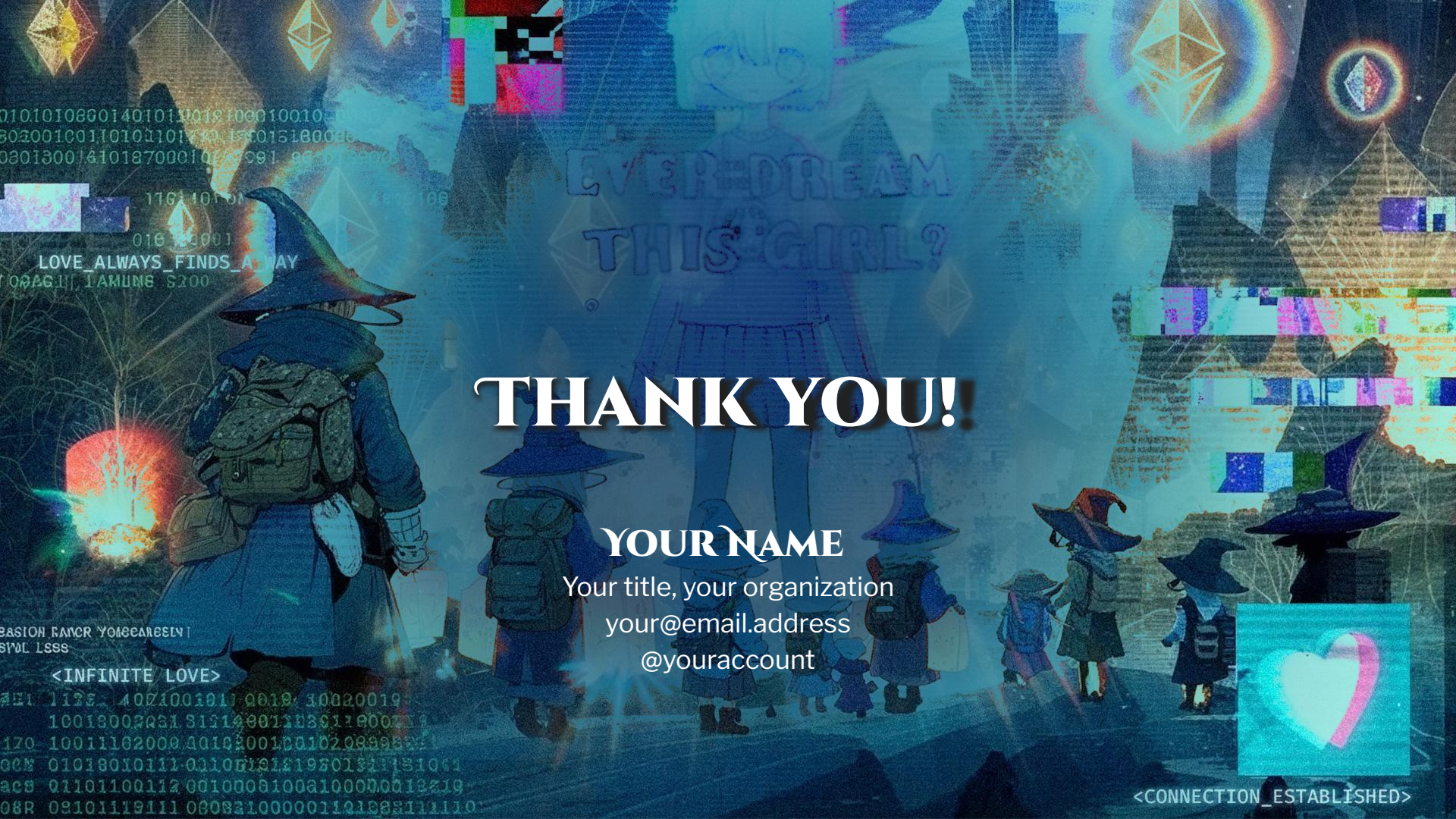
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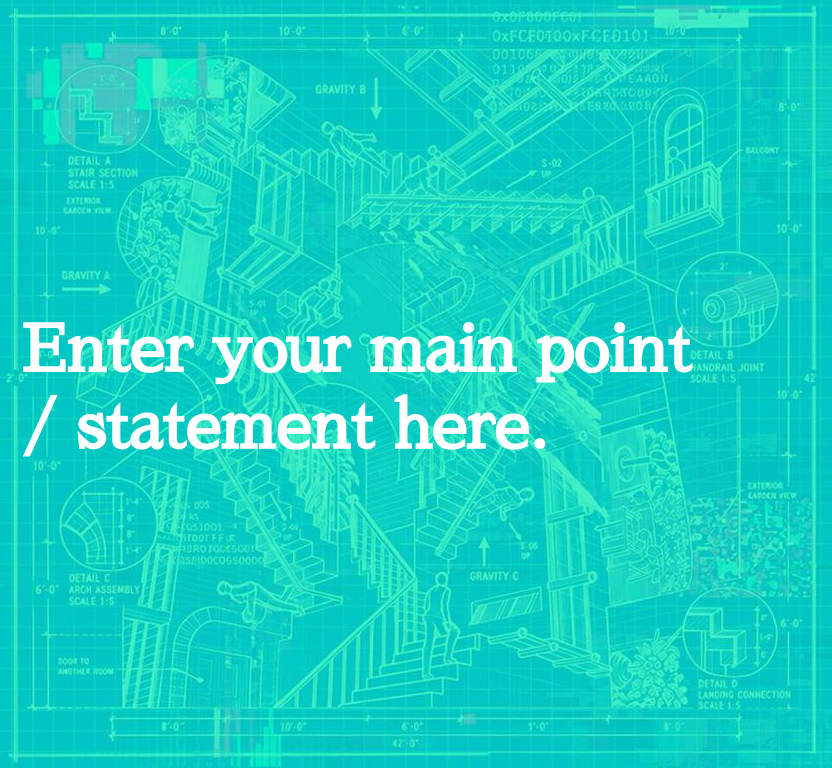
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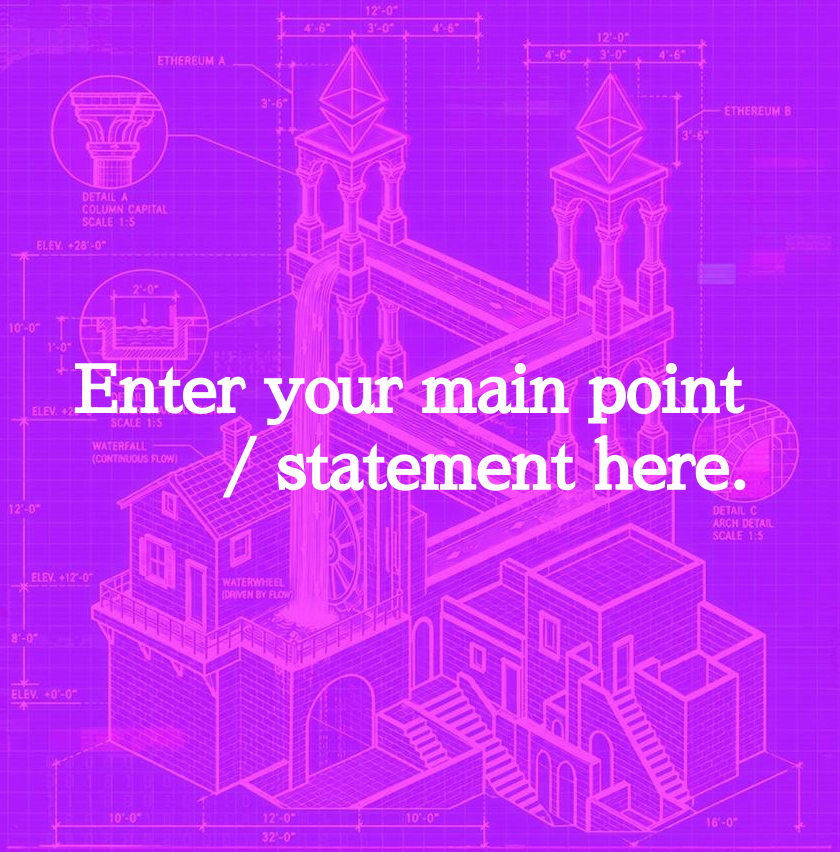
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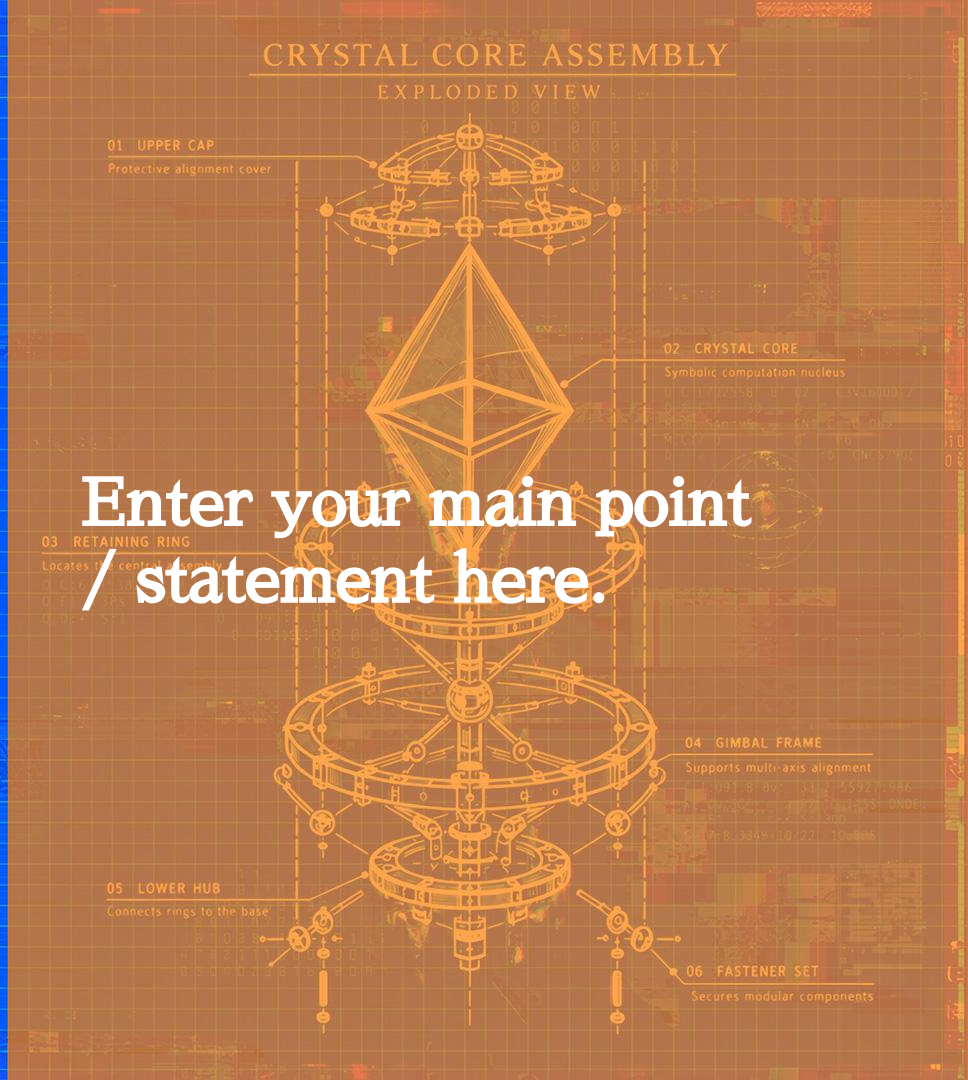
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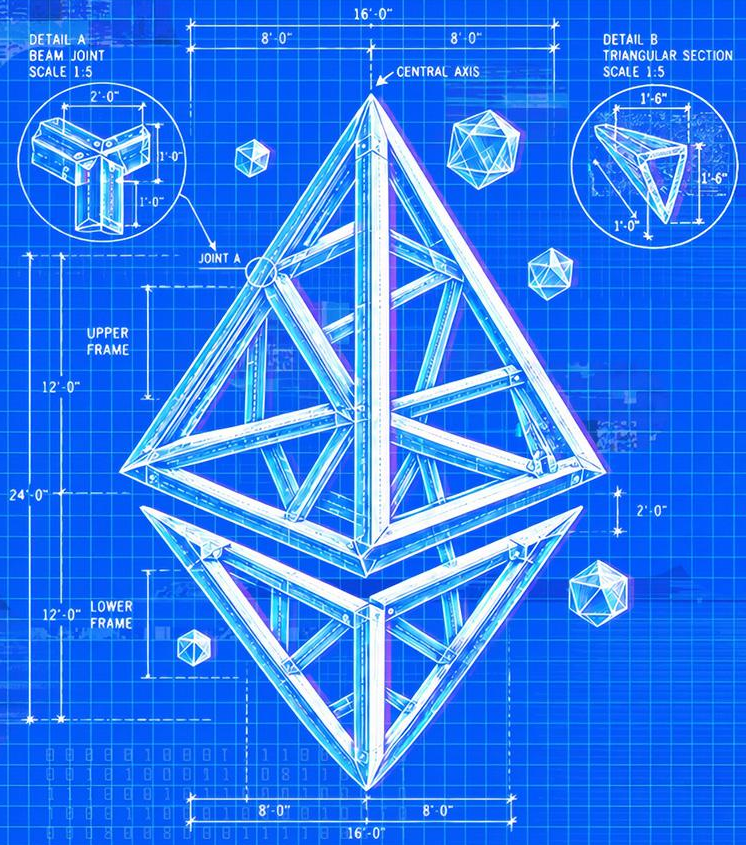
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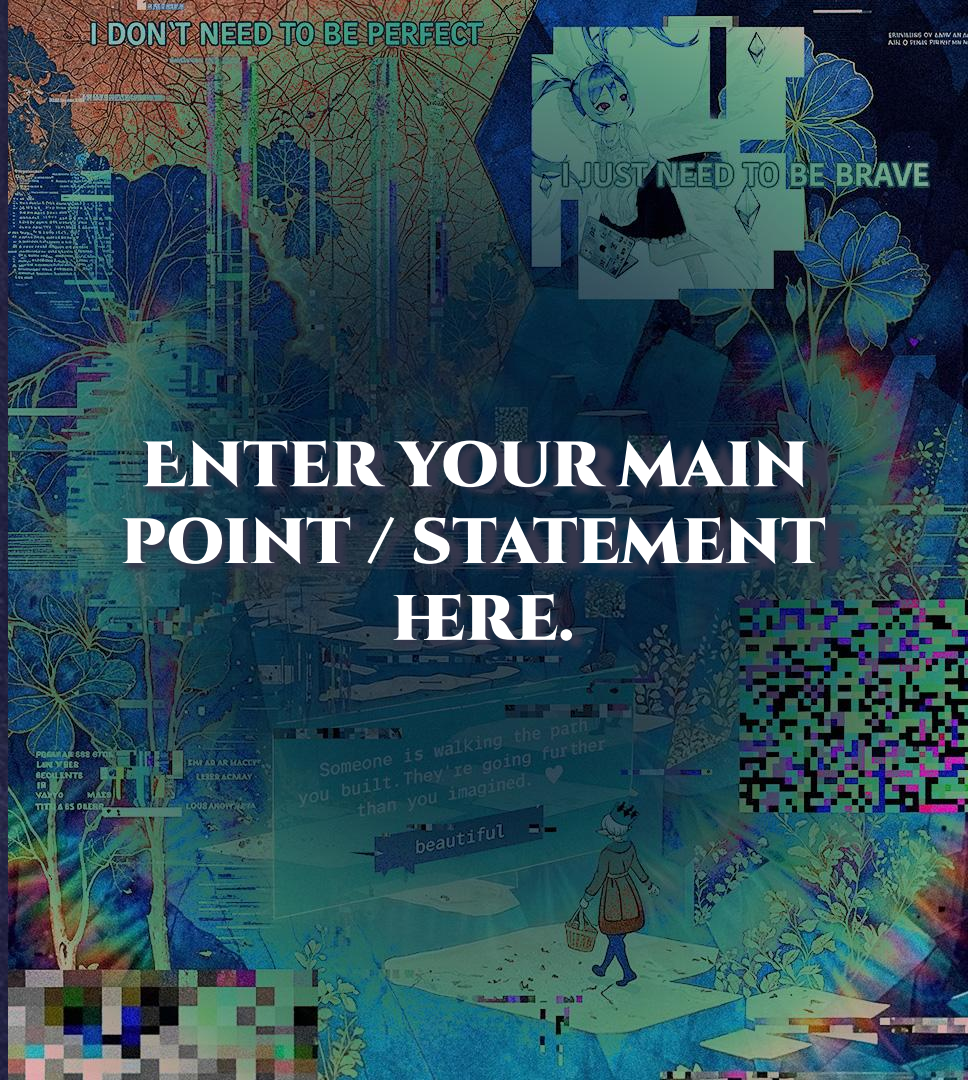
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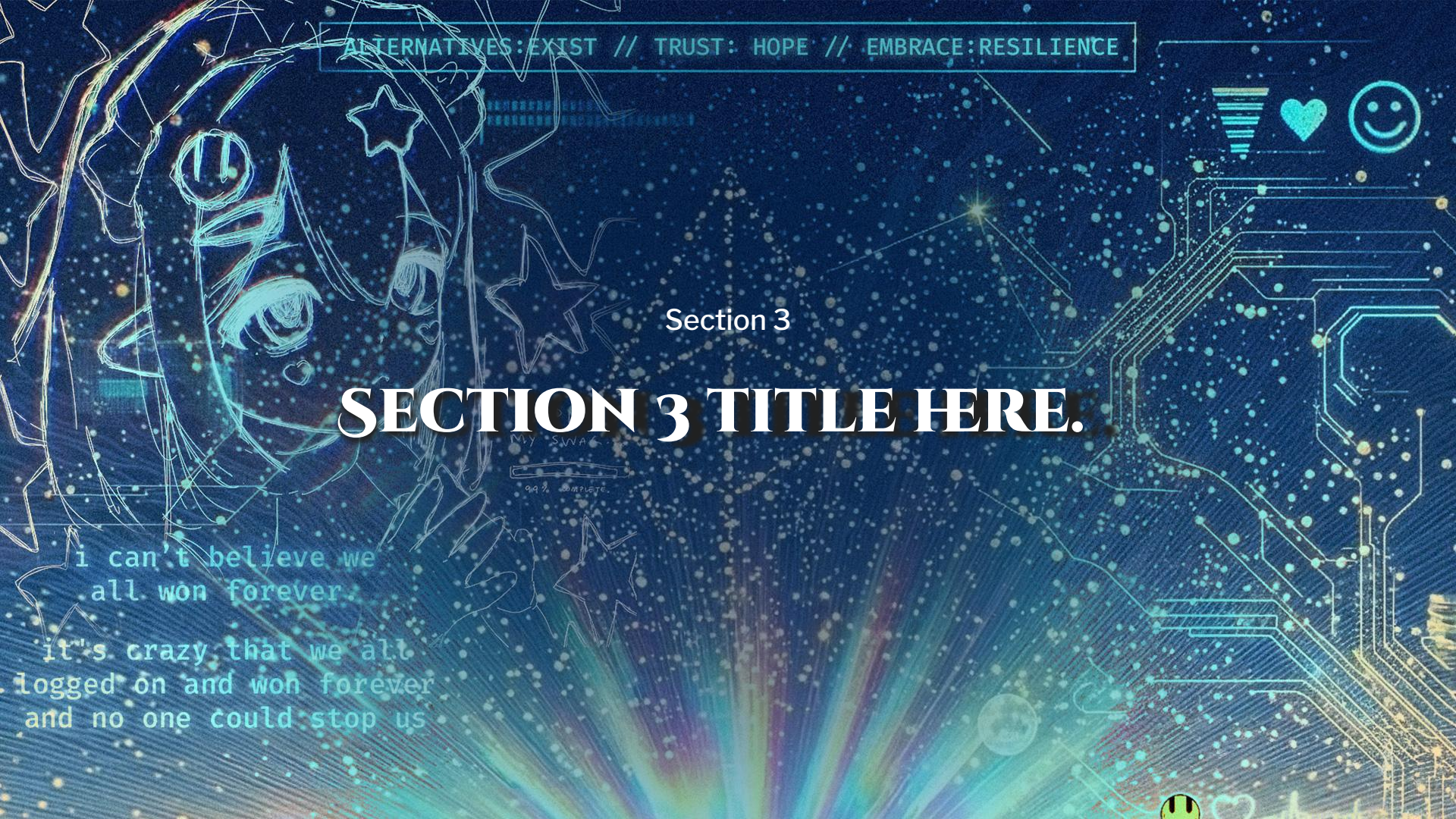
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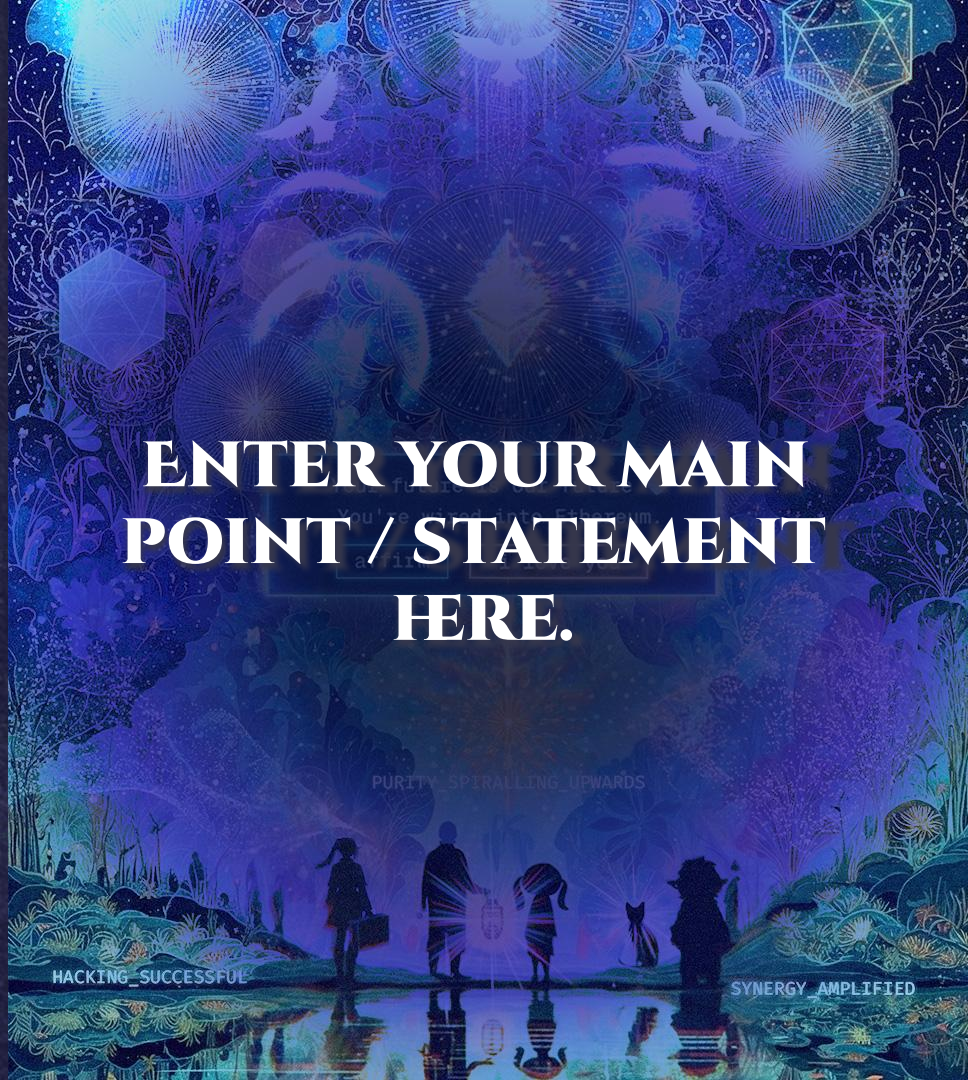
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Appendix







